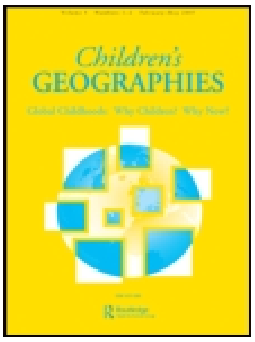

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The relationship between school ground design and intensity of physical activity

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In this study, we investigated the relationship between school ground design and children's physical activity levels. In particular, we were interested in understanding the contribution of 'green' school ground design to physical activity levels. Data for this study were collected at an elementary school in Australia and in Canada. At each school, scans of Target Areas were completed to record the students' location and intensity of physical activity, based on the System for Observing Play and Leisure Activity in Youth (SOPLAY) (Australia: 23 scans, 6 Target Areas; Canada: 18 scans, 7 Target Areas). At both schools, the highest percentage of children present was engaged in vigorous physical activity on the manufactured equipment (42% of children/scan). Similarly, at both schools, the green area encouraged the highest percentage of children present to be engaged in moderate physical activity (47% of Australian children/scan, 51% of Canadian children/scan). The patterns of sedentary behavior differed slightly between countries. At the Australian school, the paved sporting courts (57%) and the paved canteen courtyard (50.5%) promoted the highest degree of sedentary play. At the Canadian school, the treed grassy berm (42%) and the treed concrete steps (43%) encouraged the highest percentage of sedentary behavior, followed by the open asphalt (34%). These results are also discussed in light of gender distribution. We conclude with a discussion of the design and cultural factors that influence children's physical activity on school grounds. We argue that if school grounds are to realize their potential to promote physical activity, they should include a greater diversity of design features and 'green' elements that engage children of varying interests and abilities in active play.

Keywords: school ground; physical activity; natural elements; green play

Introduction

Over the past 20 years, the increasingly sedentary lifestyle of children, coupled with access to high fat, energy rich foods, has seen a marked increase in the prevalence of many diseases and adverse health conditions (Young *et al.* 2000). The most common of these are overweight and obesity, which have in turn been linked with several diseases including type 2 diabetes, heart disease, gallbladder disease and hypertension (Must *et al.* 1999). Despite the growing awareness of such trends in the childhood population, incidence of overweight and obesity have been found to be on the increase in children in most countries (Booth *et al.* 2001, 2003, He and Beynon 2006) and, as yet, no country has been able to reverse this trend (World Health Organization 1998, 2003).

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The prevalence of overweight and obesity in children has prompted many health professionals and researchers to implement several intervention strategies to tackle this epidemic. Strategies attempted on a global scale have included the recommendation of food policy changes concerning trading and marketing of low fat, high fiber foods rather than high fat, high sugar foods, and the establishment of daily physical activity guidelines applicable for all countries worldwide (World Health Organization 2003). Nationally, Australian and Canadian researchers have implemented several interventions, including the trial of several health promotion programs in schools (Active Healthy Kids Canada 2007, Australian Health Promoting Schools Association, no date) and the establishment of a national set of guidelines for physical activity in Australia (Australian Council for Health 1996) and Canada (Active Healthy Kids Canada 2007). Although this is a promising starting point, research indicates that there is still more to be done in tackling this health crisis, as the effect of several intervention strategies is still uncertain (Sallis *et al.* 2001, Zask *et al.* 2001).

In order to address the physical inactivity problem associated with rising overweight and obesity levels, it has been recommended that children spend at least 60 minutes per day in Australia (Commonwealth of Australia 2004) and 90 minutes per day in Canada (Active Healthy Kids Canada 2007) engaged in physical activity. Research has indicated that a significant number of children do not meet such guidelines, and researchers have looked at a number of ways to address this issue (Ridgers *et al.* 2006, Spinks *et al.* 2006).

Recently, the school playground has emerged as one possible site for intervention, as students spend approximately 25% of their school day in the school playground, which equates to approximately 110 minutes per day in recess breaks (Bell and Dyment 2006b). Therefore, the school playground is definitely a promising site for intervention in enabling children to reach the recommended 60 (Australia)/90 (Canada) minutes of recommended daily physical activity.

School ground interventions

In recent years, several studies concerning the manipulation of the school playground to promote physical activity have been conducted, with largely positive results. Studies have shown that interventions such as introducing game equipment (Verstraete *et al.* 2006), painting markings on the school grounds (Stratton 2000, Stratton and Mullan 2005), and introducing supervised physical activity programs (Pangrazi *et al.* 2003) can all have a positive influence on the levels of physical activity observed in children during school break times. Studies focusing on the effect of factors such as age, gender and seasonal variability on physical activity in the playground have also been conducted, with varied results (Ridgers *et al.* 2006). Despite some positive findings, it has been argued that more research is required into which of these interventions is most influential in promoting physical activity in the school playground (Ridgers *et al.* 2006).

More recently, an area of the school ground that has received attention as a possible site for intervention is the availability of green, natural settings for physical activity (Bell and Dyment 2006b, 2008, Dyment and Bell 2007a, 2007b). The process of developing such areas has been described as ‘greening’, a broad term used to encompass the efforts of students, parents, teachers, neighbourhood residents, and school and city officials who work to upgrade the physical environment and to re-establish the natural habitats that existed prior to asphalt. Some school grounds around the world are now thoughtfully designed spaces that include a variety of natural elements such as trees, butterfly gardens, ponds, and vegetable patches. A number of terms have been used to describe these changes occurring on school grounds, including ‘school ground gardening’, ‘school ground naturalization’, ‘school ground restoration’, and 'school ground greening'. While there are important differences between each term, and while each term is itself somewhat contested, for the purpose of this paper, 'school ground greening' will be used to describe collaborative efforts to improve school grounds.[1]

The greening process has become prominent in several countries worldwide, with countries including Australia, Canada, Denmark, the United States, the United Kingdom, New Zealand and South Africa embracing natural settings in the school playground (Bell and Dyment 2006b). Over the years, researchers from several disciplines have studied a variety of impacts of these spaces, including the physical, social and mental dimensions of health and well-being (Kirkby 1989, Greenwood *et al.* 1998, Barbour 1999, Malone and Tranter 2003, Bell and Dyment 2008). They have found several positive influences of green school grounds on several dimensions of health, but to date, there appears to have been limited evidence-based research directly studying the effect green school grounds have on physical activity levels of children and the wider community.

Our earlier exploratory research in Canada suggests that green school grounds, alive with trees, gardens and various other natural elements, have a significant impact on physical activity, as well as many other aspects of children's health (Bell and Dyment 2006b, 2008, Dyment and Bell 2007a, 2007b). In particular, our earlier study (Bell and Dyment 2006b), entitled *Grounds for action: Promoting physical activity through school ground greening in Canada*, examined information provided by 105 parents, teachers and administrators across 59 elementary schools in Canada, and revealed some significant key findings. One finding was that green school grounds appeal to a wider variety of student interests and support a wider variety of play opportunities that promote all levels of physical activity (Bell and Dyment 2006b). This finding alone has opened the door for further research into this area; however, other key findings have demonstrated an even wider potential for learning in greened areas. This study, as well as recent follow up studies focusing on the greening process, have also found that green school grounds promote social and cognitive well-being, create opportunities for the broader community to participate in outdoor, educational leisure activities such as gardening, and offer those students less inclined to participate in rule-bound, vigorous physical activity the chance to enjoy light to moderate physical activity through open-ended play in a natural setting (Dyment 2005, Bell and Dyment 2006b, 2008, Dyment and Bell 2006).

Despite the potential of greening that we have identified, more research is needed. We submitted the *Grounds for Action* report for evaluation to a panel comprised of experts in the fields of health and human performance, landscape architecture, nutritional science, physical education and sustainable development. The panel made several key recommendations for future research in this area (Bell and Dyment 2006a). Central to these recommendations was the need for quantitative empirical data based on direct observations of children in the school playground (Bell and Dyment 2006a). Such data were needed to confirm (or refute) our earlier finding that green school grounds encourage moderate physical activity.

The research reported in this paper seeks to fulfill the recommendation of the review panel. More specifically, it examines the effect of school ground design elements on the physical activity levels of children using direct, quantitative measurement of children's physical activity. It explores if and how the greening process can be a viable intervention to tackle physical inactivity problems such as overweight and obesity, which have reached epidemic proportions in recent years. Key research questions framed the present study:

1. How do different Target Areas promote different levels of physical activity intensity for the entire school population?
2. How does the intensity of physical activity differ between boys and girls in each Target Area?

Methods

Setting

Two primary schools – one in Australia and one in Canada – were purposefully chosen to be involved in this study.

The Australian school: The school (Kindergarten to Grade 6) was chosen because it had a diversity of play areas that allowed for an exploration of the relationship between design and physical activity. The school has just over 400 students, and is located in a middle-upper class neighborhood in Launceston, Tasmania.

The Australian school ground was divided into six Target Areas (excluding areas that were out of bounds during recess), based on key design elements. These were:

1. Green Area (large grassed area with trees, rocks, tree stumps and sandpits).
2. Manufactured Play Equipment (includes all areas with slides, metal forts, monkey bars and swings).
3. Paved Sporting Courts (fenced asphalt area designed for basketball and tennis).
4. Paved Thoroughfare (includes all concrete walkways between Target Areas).
5. Canteen Courtyard (a large asphalt area outside school canteen).
6. Mini Oval (a small, flat, dirt-surfaced recreational space).

The Canadian school: The school (Kindergarten to Grade 8) was chosen because it has been involved for 9 years in school ground greening initiatives and is seen as exemplary in the field. The school ground that originally consisted of pavement, a playing field, and two tennis courts now includes a number of 'green' design elements, including (1) a bio-diverse native species habitat and learning garden; (2) a food garden; (3) a school-wide composting system; and (4) a stone amphitheatre where classes and groups can meet outdoors. The school, with just over 700 students, is located in an upper socioeconomic community in Toronto, Ontario.

The school ground was divided into seven Target Areas (excluding areas that were out of bounds during recess), based on key design elements. These were:

1. Treed Steps (a treed concrete steps area, with seating).
2. Manufactured Play Equipment (a fixed play equipment area, with six separate pieces of equipment).
3. Green Area (a greened area, with a rock amphitheatre, food garden, native trees and shrubs).
4. Open Playing Field/Oval (an open grassy field, with a baseball diamond).
5. Treed Hill (a treed grassy berm, with picnic tables).
6. Open Asphalt Area (paved area, with four-square markings, a soccer net).
7. Paved Sporting Courts (fenced asphalt area designed for basketball, soccer).

At both schools, agreement among researchers and data collectors was established on the location, size, and boundaries of each Target Area prior to data collection.

Instrumentation

The System for Observing Play and Leisure Activity in Youth (SOPLAY) was used to track the location and intensity of play behaviors at the school population level (McKenzie *et al.* 1991, 2000, 2006). This method involved systematic, periodic scans of boys and girls in pre-determined Target Areas (see above) throughout the school ground. During each scan, each individual in each Target Area was coded as Sedentary, Moderately Active or Vigorously Active, based on momentary time-sampling. (These three codes are equivalent to SOPLAY's Sedentary, Walking and Very Active codes.) Separate scans were conducted for boys and girls.

Scans were conducted during three daily measurement periods (two at lunch hour, one at recess). The data collectors conducted a total of 23 and 18 scans of each Target Area at the Australian and Canadian school, respectively. Data collection in Australia occurred over an 11-day period in July 2007. Data collection in Canada took place over a seven-day period in April 2007.[2]

Ethics

This study received ethical clearance from both the school boards and the University of Tasmania. The principals of both schools gave their consent for participation.

Validity

Validity of testing procedures was improved as SOPLAY is a standard instrument for measuring macro-population data (McKenzie *et al.* 2000). The activity codes used in SOPLAY have been successfully used in a number of previous observation systems, and have been validated by several heart rate monitor tests (McKenzie *et al.* 2000). Dr Thomas McKenzie, the creator of SOPLAY, provided our research team with a DVD recording that trained our research team on the mechanics of using SOPLAY (McKenzie 2005). This DVD recording contained information on SOPLAY, followed by practice activities using video scenarios. The DVD was viewed a number of times by all members of the research team, and the practice activities conducted repeatedly until all researchers were confident that data from SOPLAY could be accurately recorded.

Reliability

Reliability of SOPLAY testing was improved by ensuring inter-observer reliability. According to Gratton and Jones (2004), this refers to the degree to which two or more researchers obtain similar results when assessing the same situation. To ensure that all researchers collected similar data to one another after viewing the SOPLAY instructional DVD, the researchers attended the schools used in the study for a period of two full school days prior to the first day of data collection. During this time, the researchers simultaneously undertook practice scans of each Target Area from exactly same vantage point, at exactly the same time, while students were playing in the school ground. Results were compared following each scan, and the process repeated, until all researchers were satisfied that no considerable differences remained between the classifications used by each member of the research team.

Results

Tables 1 and 2 show the distribution patterns of children at the Canadian and Australian schools, by Target Area. They describe how many children generally and girls and boys specifically were found, on average, in each Target Area during any scan. They also show the percentage of children, girls, and boys found in each Target Area as a function of the total found in the entire school ground. Tables 3 and 4 illustrate the different intensities of physical activity that were observed by the children generally and girls and boys specifically in each Target Area. Whilst the focus of this paper is on Tables 3 and 4 (i.e., intensity by Target Area and gender), it was important to include Tables 1 and 2 to have an understanding of the distribution patterns (i.e., what areas were the most populated). The remainder of this paper focuses almost entirely on Tables 3 and 4, but the reader might find it interesting to consider the results in light of Tables 1 and 2.

Table 1. Australia: The number of students (Mean, SD), by gender, in each Target Area per SOPLAY scan ($N = 23$ scans/Target Area)

	Total number of students in Target Area/scan (mean, SD) and Percentage of Total		
Target Area	Total students	Girls	Boys
Green Area	63.78 (15.26) 33%	30.26 (10.61) 34%	33.52 (9.85) 32%
Paved Sporting Courts	58.26 (10.60) 30%	13.74 (5.75) 15%	44.52 (7.99) 43%
Manufactured Equipment	22.04 (7.36) 11%	14.74 (5.90) 17%	7.30 (4.14) 7%
Canteen Courtyard	25.09 (6.55) 13%	21.22 (6.13) 24%	3.87 (3.15) 4%
Paved Thoroughfare	13.87 (5.71) 7%	8.26 (4.61) 9%	5.61 (2.84) 5%
Mini Oval	10.30 (4.29) 5%	0.52 (0.79) 1%	9.78 (4.19) 9%
Totals	193.34	88.74	104.60

Table 2. Canada: The number of students (Mean, SD), by gender, in each Target Area per SOPLAY scan ($N = 18$ scans/Target Area)

	Total number of students in Target Area/scan (mean, SD) and Percentage of Total		
Target Area	Total students	Girls	Boys
Green Area	37.06 (17.89) 14%	20.78 (10.96) 17%	16.28 (8.90) 11%
Paved Sporting Court (Tennis Court)	39.28 (6.16) 14%	9.61 (5.95) 8%	29.67 (8.75) 20%
Manufactured Equipment	26.50 (8.89) 10%	15.83 (7.34) 13%	10.67 (3.24) 7%
Open Playing Field	40.28 (22.67) 15%	14.33 (9.53) 11%	25.84 (14.66) 18%
Open Asphalt	89.11 (27.84) 33%	42.44 (13.36) 34%	46.67 (17.72) 32%
Treed Grassy Berm	24.27 (10.58) 9%	13.06 (9.14) 10%	11.22 (4.48) 8%
Treed Concrete Steps	15.88 (10.69) 6%	9.11 (8.19) 7%	6.78 (4.65) 5%
Totals	272.38	125.16	147.13

(1) How do different Target Areas promote different levels of physical activity intensity for the entire school population?

When looking at the intensity of physical activity as a function of Target Area at both the Australian (Table 3) and Canadian (Table 4) schools, the highest percentage of children present was engaged in vigorous physical activity on the manufactured equipment (42% of children at both schools). The manufactured equipment attracted 11% (Australian school) and 10% (Canadian school) of the total number of students found on the school ground (see Tables 1 and 2).

Table 3. Australia: Number of students (Mean, SD) and percentage of students in sedentary, moderate and vigorous activity per scan, by gender, in all Target Areas ($N = 23$ scans/Target Area)

	Percentage of total by gender in each scan Mean/Scan (SD)								
	Total students			Girls			Boys		
Target Area	S	M	V	S	M	V	S	M	V
Green Area	17.91 (7.95)	29.74 (9.24)	16.13 (6.02)	9.78 (6.05)	13.39 (5.76)	7.09 (3.03)	8.13 (4.09)	16.35 (7.04)	9.04 (4.71)
	28%	47%	25%	32%	44%	24%	24%	49%	27%
Paved Sporting Courts	32.35 (8.02)	17.52 (6.44)	8.39 (4.42)	8.13 (4.25)	4.00 (2.75)	1.61 (1.75)	24.22 (5.86)	13.52 (5.15)	6.78 (4.13)
	56%	30%	14%	59%	29%	12%	55%	30%	15%
Canteen Courtyard	14.91 (6.01)	7.39 (3.21)	2.78 (2.19)	13.48 (6.34)	5.35 (2.46)	2.39 (2.11)	1.43 (1.95)	2.04 (2.06)	0.39 (0.58)
	51%	39%	10%	64%	25%	11%	37%	53%	10%
Manufactured Equipment	6.56 (4.09)	6 (3.49)	9.48 (4.69)	4.43 (3.20)	3.91 (2.37)	6.39 (4.05)	2.13 (1.87)	2.09 (1.10)	3.09 (2.23)
	30%	28%	42%	30%	27%	43%	29%	29%	42%
Paved Thoroughfare	6.04 (3.57)	6.26 (4.04)	1.56 (1.75)	4.39 (3.19)	3.09 (2.56)	0.78 (1.00)	1.65 (1.56)	3.17 (5.15)	0.78 (1.17)
	44%	45%	11%	53%	37%	10%	29%	57%	14%
Mini Oval	2.87 (2.32)	4.91 (3.14)	2.52 (2.44)	0.22 (0.52)	0.13 (0.46)	0.17 (0.39)	2.65 (2.23)	4.78 (3.07)	2.35 (2.37)
	35%	37%	28%	42%	25%	33%	27%	49%	24%

Table 4. Canada: Number of students (Mean, SD) and percentage of students in sedentary, moderate and vigorous activity per scan, by gender, in all Target Areas ($N = 18$ scans/Target Area)

	Mean/Scan (SD) Percentage of total by gender in each scan								
	Total students			Girls			Boys		
Target Area	S	M	V	S	M	V	S	M	V
Green Area	12.89 (2.36) 35%	18.89 (5.87) 51%	5.28 (3.08) 14%	8.17 (8.81) 39%	9.89 (3.77) 48%	2.72 (2.14) 13%	4.72 (4.38) 29%	9.00 (5.12) 55%	2.56 (2.18) 16%
Paved Sporting Court (Tennis Court)	9.11 (4.04) 23%	17.06 (5.97) 43%	13.11 (6.98) 33%	3.28 (2.40) 34%	4.61 (3.42) 48%	1.72 (1.60) 18%	5.83 (2.92) 20%	12.44 (4.30) 42%	11.39 (5.74) 38%
Manufactured Equipment	6.94 (4.90) 26%	8.50 (4.06) 32%	11.06 (5.45) 42%	4.72 (3.20) 30%	4.83 (3.11) 31%	6.28 (4.04) 40%	2.22 (2.69) 21%	3.67 (2.28) 34%	4.78 (2.65) 45%
Open Playing Field	11.83 (8.03) 29%	17.33 (13.62) 43%	11.11 (8.68) 28%	4.89 (5.05) 34%	5.78 (6.36) 40%	3.67 (3.85) 26%	6.94 (4.88) 27%	11.56 (8.16) 45%	7.44 (5.37) 29%
Open Asphalt	30.11 (12.19) 34%	36.94 (14.19) 41%	22.06 (9.72) 25%	16.50 (7.37) 39%	17.44 (7.35) 41%	8.50 (5.92) 20%	13.61 (7.64) 29%	19.5 (8.98) 42%	13.56 (5.42) 29%
Treed Grassy Berm	10.28 (4.71) 42%	9.78 (7.77) 40%	4.22 (4.80) 17%	6.39 (4.93) 49%	4.78 (4.76) 37%	1.89 (3.31) 14%	3.89 (3.22) 35%	5.00 (3.85) 45%	2.33 (2.64) 21%
Treed Concrete Steps	6.78 (6.01) 43%	5.61 (5.49) 35%	3.50 (3.33) 22%	4.50 (4.85) 49%	3.44 (4.25) 38%	1.17 (1.54) 13%	2.28 (2.37) 34%	2.17 (2.09) 32%	2.33 (2.99) 34%

Similarly, at both schools, the green area encouraged the highest percentage of children present to be engaged in moderate physical activity (47% of Australian children, 51% of Canadian children) (Table 3 and 4). The green area attracted 33% (Australian school) and 14% (Canadian school) of the total number of students (see Tables 1 and 2). At the Australian school, the green area was the most popular area (highest percentage of students).

The patterns of sedentary behavior differed slightly between countries. At the Australian school, the Target Area that promoted the highest degree of sedentary play for all children was the paved sporting courts (second most populated target area; 30% of all children found there) (Table 1), with an average of 56% of all students in the area observed as being sedentary on average per scan at any one time (Table 3). This is closely followed by the canteen courtyard, which had an average of 51% of all students engaged in sedentary play during school recess and lunch times (Table 3).

At the Canadian school, the treed grassy berm (42%) and the treed concrete steps (43%) encouraged the highest percentage of sedentary behavior, followed by the open asphalt (34%) (which was the most populated Target Area; 33% of all children found there) (see Table 2).

(2) How does the intensity of physical activity differ between boys and girls in each Target Area?

The patterns for vigorous and moderate activity persisted across both countries when the data were examined as a function of gender (Tables 3 and 4). The manufactured equipment encouraged the highest percentage of girls and boys to be vigorously active at both schools (Australia: Girls 43%, Boys 42%; Canada: Girls 40%, Boys 45%), and the green area encouraged the highest percentage of moderate activity for girls and boys at both schools (Australia: Girls 44%, Boys 49%; Canada: Girls 48%, Boys 55%).

At the Australian school (Table 3), the most inactive areas for the female population on average were the paved sporting courts (59% of girls sedentary/scan) and the canteen courtyard (64% of girls sedentary/scan). These results are very interesting for the canteen courtyard, given that this was the second most heavily populated area for girls (most popular was the green area), yet more than half the occupants were sedentary (Table 1). The most inactive area for boys was the paved sporting courts (55% sedentary boys/scan). Again, this finding is especially interesting given that the paved sporting courts attracted the highest number of boys from all six Target Areas, yet was the most inactive site at the school (Table 1).

At the Canadian school (Table 4), the highest percentage of girls and boys engaged in sedentary behavior was found on the treed grassy berm (Girls 49%; Boys 35%) and the treed concrete steps (Girls 49%; Boys 34%). On the open asphalt, which attracted the highest number of girls and boys (Table 2), 39% of girls and 29% of boys were engaged in sedentary behavior.

Discussion

The findings that emerged across the Australian and Canadian schools are strikingly similar with respect to the patterns of intensity of physical activity as a function of design. At both schools, the highest percentage of children engaged in vigorous and moderate activity was found on the manufactured equipment and green areas, respectively. At the Australian school, the highest percentage of children was sedentary on the canteen courtyard (girls) and paved sporting equipment (boys); whereas at the Canadian school, the highest percentage of children were engaged in sedentary behavior on the treed grassy berm and steps (girls and boys) and open asphalt (girls). We now turn to a discussion of the design and cultural factors that might explain why children are engaged in vigorous, moderate and sedentary activity across the various Target Areas at the Australian and Canadian schools.

School ground spaces that promote vigorous physical activity

At both schools, the manufactured equipment provided opportunities for a large percentage of children to be engaged in vigorous physical activity. The contention that manufactured equipment encourages children to engage in vigorous physical activity is supported by previous findings within the literature, which have highlighted the effect of external stimuli in raising physical activity levels of children (Stratton 2000, Stratton and Mullan 2005, Verstraete *et al.* 2006). In these studies, external stimuli such as painted markings on barren surfaces (Stratton 2000, Stratton and Mullan 2005) and the introduction of game equipment (Verstraete *et al.* 2006) have all been shown to increase physical activity levels in the school ground. The common finding to all these studies is that by adding colour, open-ended play opportunities and external stimuli for imaginative play, physical activity in children within the school playground may be improved. Therefore, when analyzing the results from this study, one possible reason as to why the manufactured equipment may have promoted higher intensities of physical activity is the wealth of external stimuli present through the equipment, whereas the paved sporting courts, open asphalt, and the canteen courtyard, which were almost entirely barren and unnatural, may have promoted less activity due to the lack of external stimuli (these Target Areas will be discussed below).

The manufactured equipment was a particularly important area for girls to engage in vigorous activity. There are a number of possible reasons for this observation, all of which support previous findings from a number of studies focused on gender differences and school ground design (Blatchford *et al.* 2003, Stratton and Mullan 2005, Verstraete *et al.* 2006, Paechter and Clark 2007). Blatchford *et al.* (2003), for example, found that boys were far more active than girls in those areas that promote sport based, rule bound physical activity. It may thus be suggested that one reason for girls' increased activity on the manufactured play equipment is that these areas promote open ended play, not centred on rules or sport based activity.

Another possible reason for the observed high percentage of girls engaged in activity in the manufactured equipment and the green area is the chance to engage in social interaction with other students in a non-competitive context. As suggested by Renold (1997), girls tend to play in those areas not dominated by sport based physical activity, and tend to choose social interaction over competition in the school playground. We observed the manufactured equipment to provide a non-competitive environment in our study, and repeatedly found that girls were dominant to boys in the average number of students in this area (see Tables 1 and 2).

The paved sporting courts in Canada did promote a large percentage of boys (38%) to engage in vigorous physical activity (Table 4). This was also the second most popular area for boys after the open asphalt (20% of all boys in the school ground were found there, on average, during any scan) (Table 2). These were actually tennis courts, with clearly defined boundaries (chain link fences) that were dominated by active competitive, rule-bound games (e.g., soccer, octopus) involving lots of boys at any one time. Few girls were found on the courts (8% of total), and of these, 34% were engaged in sedentary behavior (often seen at the periphery of the courts) (Table 4 and Table 2).

Whilst the paved sporting courts in Australia were the most popular Target Area for boys (43% of all boys on the school ground were found there, on average, during any scan) (Table 1), the intensity observed was quite different from the Canadian school. We found a notable lack of vigorous physical activity on the paved sporting courts, with 55% of all boys, on average, being deemed as sedentary (Table 3). Little evidence appeared to be available to support this finding in the existing literature, although observations made during data collection may provide an insight into why this area promoted such a high percentage of boys to be inactive. The games most frequently observed on the Australian paved sporting courts were elimination style games, involving, for example, long lines of students and only two people shooting a basketball at a time, and a soccer style game involving the ball being kicked between large groups of people against a wall, again only providing opportunities for one or two boys to be moving at a time. It appears that these games have been developed to allow inclusion of large numbers of people, but do not promote physical activity for most people within this Target Area. Perhaps a way of increasing activity on the Australian courts would be to provide more balls, more basketball hoops (Verstraete *et al.* 2006), or more painted goal markings on walls and fences (Stratton 2000, Stratton and Mullan 2005) to allow more than one game to be in progress at any one time and more children to be involved at higher intensities.

The finding that the paved sporting courts promoted rule bound physical activities that were dominated by boys is unsurprising. Blatchford *et al.* (2003) found that boys tend to dominate those areas within the school playground that are designed for ball sports, while girls tend to be less interested in sport based physical activity. Similar results were found by Renold (1997), who found that girls tended to situate themselves around the perimeter of sporting areas or in other playground areas, and were more likely to desire socialization rather than competition. Furthermore, Paechter and Clark (2007) suggest that physical activity in the school playground is largely affected by socialization, and that the social expectation of children in western society, particularly for boys, is to be active through competitive sport.

School ground spaces that promote moderate activity

At both the Australian and Canadian schools, the green area of the school ground promoted the highest percentage of both boys and girls to be engaged in moderate physical activity. In our study, we observed Australian and Canadian students in the green area spending recess and lunch breaks exploring areas hidden by trees and bushes, climbing over boulders, playing in the sandpit and crawling through long grass in imaginative play, all of which were deemed as moderate physical activity. This finding confirms that of our earlier exploratory survey in which the majority of the 151 respondents (71%) indicated that school ground greening had resulted in more moderate and/or light physical activity on their school grounds (Bell and Dyment 2006b, 2008).

These findings certainly challenge the belief that flat turf and asphalt provide ideal (and often only) surfaces for burning off excess energy and engaging in team sports, and are therefore best suited to promoting physical activity.

These findings may have important implications for health promoters seeking to provide moderate physical activity alternatives for children not interested in vigorous competitive play or sport based physical activity. As suggested in related literature, moderate physical activity (as opposed to vigorous physical activity) is increasingly being seen to be sufficient in targeting overweight and obesity in children (Frank and Niece 2005). Moderate physical activity that offers the chance for enjoyable, non-competitive physical activity is vitally important for reaching out to those children who do not enjoy traditional forms of vigorous sport based physical activity. With this in mind, green school ground spaces can be promoted as a highly effective intervention to childhood physical activity and associated health risks, especially for girls, who have repeatedly been found to be less active in the school playground than boys (Pangrazi *et al.* 2003, Ridgers and Stratton 2005, Verstraete *et al.* 2006).

The green area of the school ground seems to allow children to 'expand their play repertoire' (Moore and Wong 1997, p. 91) by engaging them in less organized play and more unorganized 'free' play. On the green school ground in Berkely, CA, Moore and Wong observed an increase in active play, creative play, pretend play, exploratory play, constructive play and social play as compared to the original school ground. They noted,

> This was a far cry from the old school ground, where girls hung around admiring the boys' prowess at playing ball or felt excluded because they were not attracted by the crowded play equipment; and where nonathletic children were ridiculed for not participating in the unchanging routines of ball courts, game lines, and metal bars. (p. 91)

Indeed, evidence suggests that children desire natural, complex, challenging and exciting play environments that provide options and choice for play (Moore and Wong 1997, Stine 1997). In light of this desire, it is not surprising that green school grounds appeal to a wider variety of student interests, support a wider variety of play activities and promote moderate physical activity (Dyment and Bell 2007b). Indeed, many researchers have documented the changes in children's play behaviors as a result of greening, noting in particular an increase in the diversity of play behaviors (Faber-Taylor *et al.* 1998, Tranter and Malone 2004). On green school grounds, trees, shrubs, rocks and logs define a variety of places to jump, climb, run, hide and socialize. Moveable, natural materials such as sticks, branches, leaves and stones provide endless opportunities to engage in imaginative play, such as building shelters and huts – an appealing and almost universal experience of childhood that gets them moving (Cobb 1977, Sobel 1993).

A study from Sweden supports these findings, indicating that the physical qualities of outdoor preschool environments (their size, the presence of trees and shrubs, the proximity of play structures to vegetation) are an important trigger of physical activity (Boldemann *et al.* 2006). Using pedometry to measure and compare children's movement at 11 different preschools, the Swedish researchers found that children were taking a significantly higher number of steps in spacious play environments with trees, shrubbery and broken ground. Seen in this light, green school areas stand to be an important intervention to promote physical activity.

Similarly, an American study investigating the impact of design on children's physical activity at three childcare centres indicates that the size and diversity of the play areas are among the most important variables predicting physical activity (Cosco 2006). Using accelerometry, behaviour mapping and video-tracking of individual children, Nilda Cosco found that the most active play area among the three schools contained vegetation, pathways and manufactured items.

The diversity of natural features on school grounds can also have a positive influence on motor development. Ingunn Fjortoft (2004) compared the physical fitness of 5-, 6- and 7-year-old children playing in a natural playscape (a forest) adjacent to their school and children playing in a more conventional playground. Those playing in the natural playscape showed a statistically significant increase in motor fitness and showed greater improvements in balance and fitness than those in the other group. According to Fjortoft, landscape features 'influence physical activity and motor development in children' (p. 21) and play activities 'increase with the complexity of the environment and the opportunities for play' (p. 24).

The green area of the school ground may be an especially important space for children who are less physically competent or socially developed (due to age, gender, ability) to engage in moderate physical activity. American researcher Ann Barbour (1999) sheds some insight in this regard in her study that compared play behaviours on two school grounds: one that provided primarily opportunities for physical play, and another that provided for a diversity of play opportunities. At schools that only provided opportunities for active and physical play, social hierarchies were established through these means, and children with low physical competence or desires were often socially excluded. Conversely, at schools where a diversity of play opportunities were afforded, students who were less physically competent could still engage in types of physical activity that were more in line with their abilities and interests.

School ground spaces that promote sedentary behavior

There were several Target Areas at the schools that promoted limited opportunities for physical activity. At the Australian school, for example, more than half of the girls on the canteen courtyard were sedentary (second most popular area for girls) and just over half of the boys were sedentary on the paved sporting courts (most popular area for boys). At the Canadian school, a high percentage of girls and boys were inactive on the treed grassy berm, the treed steps, and the green area. A high percentage of girls was also inactive on the paved sporting courts, the open playing field, and the open asphalt.

On one hand, these findings can be seen as problematic. Why are so many children sedentary on the paved sporting courts, open asphalt and open playing field? Aren't these spaces, that so often dominate school ground settings (Paechter and Clark 2007), deemed to be ideal spaces for allowing children to 'burn off steam'? (Evans 1997). Whilst this is a commonly held notion, our findings suggest a different story.

It appears that these traditional sporting spaces at both schools are often dominated by boys – in particular, physically competent boys that dominate and control the play spaces. As a result, girls and less physically competent children are often relegated to the sidelines of these sporting spaces, unable or unwilling to participate in the dominant activities (Barbour 1999). Paechter and Clarke (2007) note that 'girls are often systematically excluded by boys from much of the space of the playground' (p. 320). They explain that 'the gendered dominance of the playground by boys and their use of space consuming activity games such as football to construct and confirm their masculinities has concomitant effects on the possible activities of girls' (p. 320). Yet these sporting spaces often dominate the spatial area of most school grounds, leaving few opportunities for girls and other less physically competent children to engage in physical activity. As a result, important opportunities for encouraging children to engage in physical activity are lost.

The barren design of these sporting spaces might also explain why so many children are inactive. Recent research points to interventions such as the introduction of game equipment (Verstraete *et al.* 2006), playground markings (Stratton and Mullan 2005), and a diversity of features (Moore 1989) that might encourage more activity.

On the other hand, the provision of spaces that promote sedentary activity should not be seen as entirely problematic. Recess and lunch hour times fulfill many 'roles' in the lives of boys and girls, beyond the commonly held one of simply burning off steam. Evans (1995) found boys and girls have quite different attitudes towards their break times during school, with boys viewing recess and lunch breaks as an opportunity to be active and play competitive games, whereas girls view these times as a chance to talk with friends and socialize. Observations from our study strongly support these findings, as boys on the sporting areas were generally playing competitive games of basketball and soccer, whereas the canteen courtyard, treed grassy berm and steps were often used by large numbers of girls who chose to simply sit in circles and talk, play quiet games, or engage in creative games and generally play in a manner that was more nurturing, more cooperative and less competitive.

Conclusions and recommendations

This study adds to a growing body of research that is exploring the role of school grounds and recess time in promoting children's physical activity (Stratton 2000, Stratton and Mullan 2005, Bell and Dyment 2006b, Ridgers *et al.* 2006, Verstraete *et al.* 2006, Paechter and Clark 2007). Strategic interventions in school ground design are needed in order to provide opportunities for boys and girls to be more physically active during the school day. Conventionally designed school grounds, that consist primarily of open expanses of turf and asphalt, offer valuable opportunities for vigorous activities in rule-bound games like basketball, tag and soccer. Our findings suggest, however, that conventional school grounds have their limitations in promoting physical activity because many children are not interested or able to play in such vigorous games. In such cases, important opportunities for engaging in daily physical activity are lost.

This research has also added to our understanding of how gender is a prominent factor influencing health disparities generally and physical activity in particular. Research indicates that boys tend to be more physically active than girls (Action for Healthy Kids 2004, Ridgers *et al.* 2005, Active Healthy Kids Canada 2007), and several researchers have noted the different play behaviours of boys and girls throughout a number of developmental stages (Hart 1987, Moore 1986, Nabhan and Trimble 1994, Cunningham and Jones 1996, Gagen 2000). Some have argued that the design of outdoor play spaces must be taken into account to enable girls 'to construct their femininities around active and assertive play' (Paechter and Clark 2007, p. 329). Our study has shown that the majority of conventional school grounds that are comprised of asphalt and open playing field do little to promote moderate or vigorous physical activity for many girls. On these spaces, girls are often engaged in sedentary behaviour, unable, unwilling or uninterested in playing competitive rule-bound games like soccer and football. There is thus a critical need to provide spaces that will engage more girls in moderate physical activities. Greened areas stand to make an important contribution in this regard. In this study, almost half the girls (44% in Australia; 48% in Canada) were engaged in moderate physical activity in the greened area. While it is important not to reinforce simplistic gender stereotypes (there are, of course, girls who want to run and play competitive games and boys who want to engage in quieter activities), the findings from our study highlight the value of greened areas that can accommodate a range of active and quiet, competitive and cooperative, rule-bound and open-ended play activities.

If school grounds are to realize their potential to promote physical activity, they must offer options for active play that appeal more broadly to children of varying interests and abilities. This is where green design elements stand to make an important contribution and get more children moving in ways that nurture all aspects of their health and development.

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Notes

1. We are not suggesting that these terms (i.e., school ground gardening, school ground naturalization, school ground restoration, school ground improvement, school ground greening) all mean the same thing. Nor are we suggesting that debate about their definitions is not worthy. Such debate is, however, not the main focus of this paper. For a more detailed explanation of the differences between each term, see Houghton (2003).
2. We recognize that the timing of the data collection stands to influence the findings considerably. For example, in Canada, we collected the data in April (Spring), a month characterized by cool windy days. In Australia, we collected the data in July (Winter), a month noted for wet chilly days. At both research sites, we collected data only on somewhat fine days (i.e., where the students were allowed to play outside during recess/lunch). We do, however, recognize, that our seasonal choice influences the findings (for example, on hot summer days, the cool shady areas of the school ground might be more in demand in both countries whereas this was not an issue in our study).

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